



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL
2010

ASSESSMENT TASK 1

Mathematics
Extension 2

General Instructions

- Working time – 50 minutes
- Write in the booklet provided on both sides of the paper
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
☐ Mr Trenwith
☐ Mr Weiss

Student Name: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total
Mark	9	13	15	12	49	100

Question 1 (Start a new page)

- a) If P represents the complex number z , sketch the locus of P if

$$\operatorname{Im}(z) > 1 \text{ and } 0 \leq \arg z \leq \frac{\pi}{4} \quad 2$$

- b) Given $w = \frac{2-3i}{1+i}$. Find: 3

i) $|w|$

ii) $\bar{w}$

iii) $w + \bar{w}$

- c) Given $z = x + iy$, prove that $z + \bar{z} = 2\operatorname{Re}(z)$ 2

- d) Sketch the locus of $\arg\left(\frac{z-1}{z+2i}\right) = \frac{\pi}{3}$ 2

Question 2 (Start a new page)

- a) Prove that if u and v are real the following is purely imaginary 2

$$\frac{u + iv}{u - iv} - \frac{u - iv}{u + iv}$$

- b) Find the modulus and argument of $z = \left(\frac{1+i}{1-i}\right)^2$ and express the answer in modulus-argument form 3

- c) i) Find (a, b) such that $(3+2i)(a+ib) = 7-5i$ 3

ii) Find $\sqrt{5-12i}$ 2

- d) If $z_1 = 2 + 5i$ 3

$$z_2 = 3+4i$$

express z_1 , z_2 , and $z_1 z_2$ in modulus-argument form.

Examination continues on next page ...

Question 3 (Start a new page)

- a) Find the real roots of $x^4 + 2x^3 - 12x^2 + 14x - 5 = 0$, given that it has a triple root. **3**
- b) Given that $2 - 3i$ is a zero of $P(x) = x^3 + x^2 - 7x + 65$, find the other zeros. **3**
- c) Solve $x^3 - 3x^2 - 6x + 8 = 0$ given that the roots are in geometric progression. **3**
- d) If w is a non-real cube root of unity (i.e. $w^3 = 1$)
- i) Prove $\frac{1}{1+w} + \frac{1}{1+w^2} = 1$ **3**
- ii) Show $\frac{1+w^n+w^{2n}}{3} = 1$ or 0 . According to whether n is or is not a multiple of 3. **3**

Examination continues on next page ...

Question 4 (Start a new page)

a) Let $z = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$

i) Find z^6 . 1

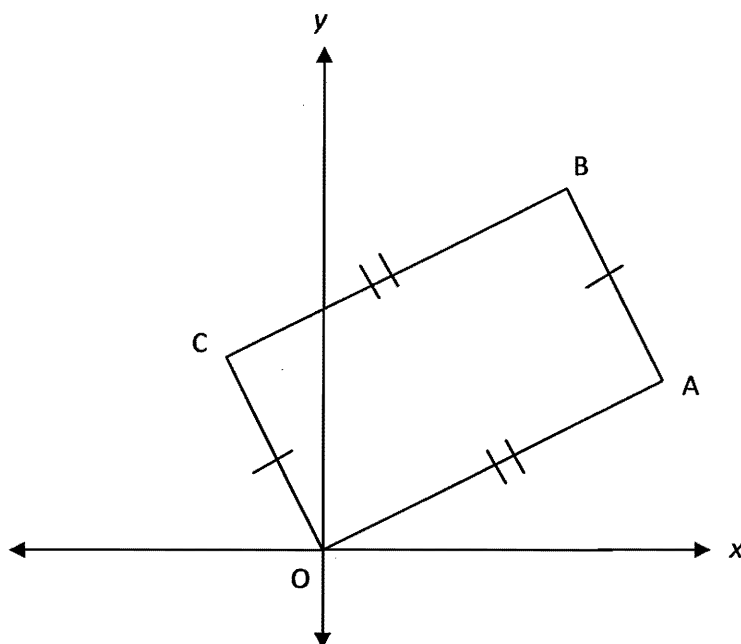
ii) Plot on an Argand diagram, all complex numbers that are solutions of $z^6 = -1$ 2

iii) Hence or otherwise, find the roots of $z^4 + z^2 + 1 = 0$ 3

b) Sketch the region on the Argand diagram that satisfies the inequality

$$z\bar{z} + 2(z + \bar{z}) \leq 0 \quad \text{3}$$

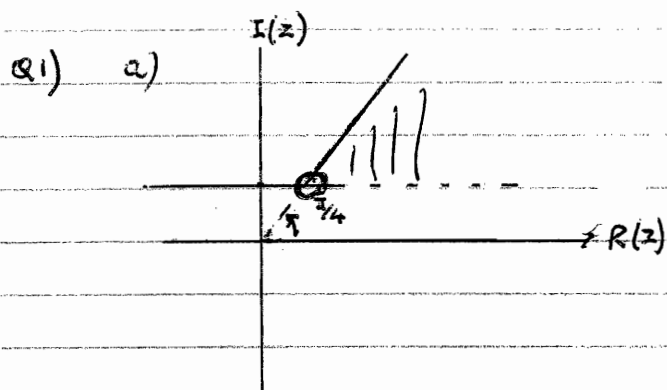
c) In the Argand diagram $OABC$ is a rectangle where $OC = 2OA$. The vertex A corresponds to the complex number w



i) What is the complex number that corresponds to the vertex C . 1

ii) What complex number corresponds to the point of intersection D of the diagonals OB and AC . 2

End of examination.



2

1 left for
each thing.
Missing
No
open circle
loose 1 mark

b) i/
$$W = \frac{2-3i}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{-1-5i}{2}$$

1

ii/
$$|W| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{5}{2}\right)^2}$$

$$= \frac{1}{2} \sqrt{26} = \sqrt{\frac{13}{2}}$$

1

iii/
$$\bar{W} = -\frac{1}{2} + \frac{5i}{2}$$

$$W + \bar{W} = \frac{-1}{2} - \frac{5i}{2} + \frac{-1}{2} + \frac{5i}{2}$$

$$= -1$$

1

c)

$$z = x + iy$$

$$z + \bar{z} = x + iy + x - iy$$

$$= 2x$$

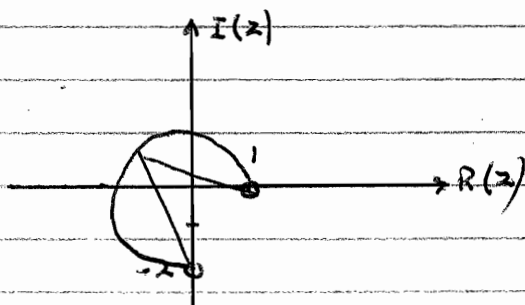
Now $R(z) = x$

$$2 R(z) = 2x$$

$$\therefore 2 R(z) = z + \bar{z}$$

2

d)



2

Q2) a) $\frac{u+iv}{u-iv} - \frac{u-iv}{u+iv}$

$$= \frac{(u+iv)^2 - (u-iv)^2}{u^2+v^2}$$

$$= \frac{u^2 + 2iuv - v^2 - u^2 + 2iuv + v^2}{u^2+v^2}$$

$$= \frac{4iuv}{u^2+v^2}$$

$$= \left(\frac{4uv}{u^2+v^2} \right) i$$

hence expression is purely imaginary

b) $z = \left(\frac{1+i}{1-i} \right)^2$

$$= \frac{1+2i-1}{1-2i-1}$$

$$= -1$$

$$|z| = 1$$

$$\arg z = \pi$$

$$\therefore z = 1 \cos \pi$$

c) if $(3+2i)(a+ib) = 7-5i$

$$3a - 2b = 7$$

$$3b + 2a = -5$$

$$6a - 4b = 14$$

$$6a + 9b = -15$$

$$13b = -29$$

$$b = -\frac{29}{13}$$

$$\therefore a = \frac{11}{13}$$

ii/

$$x+iy = \sqrt{5-12i}$$

$$x^2 - y^2 = 5 \quad 2xy = -12$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2-9)(x^2+4) = 0$$

$$\therefore \sqrt{5-12i} = \pm(3-2i) \text{ each}$$

d) $z_1 = 2+5i$

$$|z_1| = \sqrt{29}$$

$$\arg z_1 = 68^\circ 12' \text{ or } 1.19 \text{ or } \tan^{-1} \frac{5}{2}$$

$$z_2 = 3+4i$$

$$|z_2| = 5$$

$$\arg z_2 = 53^\circ 8' \text{ or } 0.93 \text{ or } \tan^{-1} \frac{4}{3}$$

$$|z_1 z_2| = |(2+5i)(3+4i)|$$

$$= |6-20+15i+8i|$$

$$= |-14+23i|$$

$$|z_1 z_2| = \sqrt{725} = 5\sqrt{29}$$

$$\arg z_1 z_2 = \tan^{-1} \left(\frac{-23}{14} \right)$$

$$= 121^\circ 20'$$

$$\text{or } 2.12 \text{ or } \tan^{-1} \frac{14}{-23}$$

$$\text{accept } \pm \sqrt{13} \cos \left(\frac{\tan^{-1} \left(\frac{12}{5} \right)}{2} \right)$$

$$= \pm \sqrt{13} \cos (-33^\circ)$$

Q3) a) $P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$
 $P'(x) = 4x^3 + 6x^2 - 24x + 14$
 $P''(x) = 12x^2 + 12x - 24$
 $P''(x) = 0$

$$12(x+2)(x-1) = 0$$

$$\therefore x = -1 \text{ or } -2$$

$$P(-1) = 1 + 2 - 12 + 14 - 5 = 0 \quad \checkmark \text{ method for next root.}$$

$$\therefore P(x) = (x-1)^3(x+5)$$

Roots are 1 (triple) and 5 $\checkmark$

b) If $2-3i$ is a root
 $2+3i$ is also a root $\checkmark$

$$\text{Now } (2-3i)(2+3i) = 4+9 = 13 \quad \checkmark$$

$$\therefore \text{ other zeros } = \frac{-6 \pm \sqrt{36-13}}{13} = -5 \quad \checkmark$$

$$\therefore \text{ Zeros are } -5, 2+3i, 2-3i$$

c) $\frac{a}{r} \times a \times ar = a^3 = -8$

$$\therefore a = -2 \quad \checkmark$$

$$\frac{a}{r} + a + ar = 3$$

$$\frac{-2}{r} - 2 - 2r = 3$$

$$\frac{-2}{r} - 2r = 5$$

$$\therefore 2r^2 + 5r + 2 = 0$$

$$(2r+1)(r+2) = 0$$

$$\therefore r = -\frac{1}{2} \text{ or } -2$$

$$\therefore a = -2, r = -2 \Rightarrow 1, -2, 4 \quad \checkmark \checkmark$$

$$a = -2, r = \frac{1}{2} \Rightarrow 4, -2, 1$$

} Roots are 1, -2, 4.

Marking for c: Answers only for this with wrong method 1 mark only.

$\checkmark$ for getting the 1st root with series correctly.

$\checkmark \checkmark$ for the other 2 roots obtained with a correct method.

d) i) $\frac{1}{1+\omega} + \frac{1}{1+\omega^2} = 1$

$$\text{L.H.S.} = \frac{1+\omega^2+1+\omega}{(1+\omega)(1+\omega^2)} \quad \checkmark$$

$$= \frac{1+\omega+\omega^2+1}{1+\omega^2+\omega+\omega^3} \quad \checkmark$$

$$= \frac{1+\omega+\omega^2+1}{1+\omega^2+\omega+1} \quad \checkmark$$

$$= 1$$

$$= \text{R.H.S.}$$

ii) $\frac{1+\omega^n+\omega^{2n}}{3}$

Let $n = 3k$ (a multiple of 3)

$$\frac{1+\omega^{3k}+\omega^{6k}}{3} = \frac{1+1+1}{3} = 1 \quad \checkmark$$

ii) $n = 3k+1$

$$\frac{1+\omega^{3k+1}+\omega^{6k+2}}{3} = \frac{1+\omega+\omega^2}{3}$$

$$= 0 \quad \checkmark$$

$n = 3k+2$

$$\frac{1+\omega^{3k+2}+\omega^{6k+4}}{3} = \frac{1+\omega^2+\omega}{3}$$

$$= 0 \quad \checkmark$$

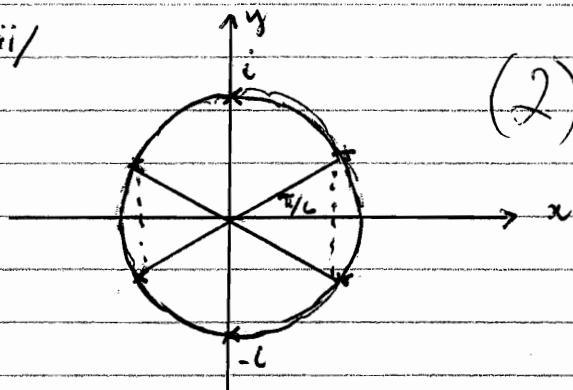
$\therefore$ Statement true

Q4) a) i/ $z = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$

$$z^6 = \cos \pi + i \sin \pi$$

$$= -1 \quad (1)$$

ii/



(2)

c) i/ $OA = w$

(1) $\therefore OC = \frac{1}{2}iw$

ii/ $OB = OA + OC$

$$= w + \frac{1}{2}iw \quad w + \frac{1}{2}iw$$

(1)

$$OD = \frac{OB}{2}$$

(1)

$$= \frac{w}{2} + \frac{iw}{4} \quad \frac{w}{2} + \frac{iw}{4}$$

iii) $z^6 - 1 = (z^2 - 1)(z^4 + z^2 + 1)$

b) $z\bar{z} = |z|^2 = x^2 + y^2$

$$z + \bar{z} = 2 \operatorname{Re}(z) = 2x \quad \text{Method}$$

(1)

$$z\bar{z} + 2(z + \bar{z}) \leq 0$$

$$x^2 + y^2 + 4x \leq 0$$

$$(x+2)^2 + y^2 \leq 4 \quad (1)$$

roots of $z^4 + z^2 + 1 = 0$

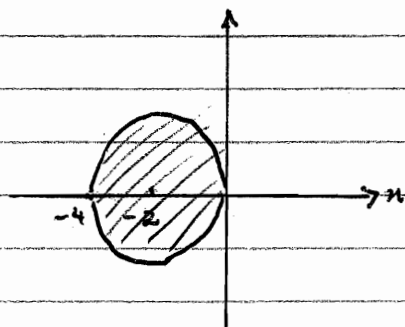
$$z^2 = \frac{-1 \pm \sqrt{1-4}}{2}$$

(1) $= \frac{-1 \pm \sqrt{3}i}{2}$

$\therefore$ roots of z

$$\frac{-1 \pm \sqrt{3}i}{2}, \frac{\pm 1 \pm \sqrt{3}i}{2}$$

(2) $\operatorname{cis}\left(\frac{\pm \pi}{3}\right) \operatorname{cis}\left(\frac{\pm 2\pi}{3}\right)$



(1)